MD FAIZ KHAN

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SUMMARY

AI/ML Engineer (fresher) with hands-on, end-to-end experience building and deploying Machine Learning, Deep Learning, and Generative AI systems – from data preprocessing and model development through evaluation and production deployment. Proficient in Python, NumPy, Pandas, Scikit-Learn, PyTorch, React, and SQL, with practical experience in Computer Vision, NLP, LLMs, Retrieval-Augmented Generation (RAG), and multi-API AI integrations (Gemini, Tavily, OpenAI). Validated modeling skills through a competitive ML leaderboard (top 9% of 1,800+ participants). Comfortable working independently, deploying full-stack applications, and communicating technical findings clearly.

EDUCATION

Indian Institute of Technology Madras Chennai, India
B.S. in Data Science and Programming | CGPA: 7.8/10 *Expected 2027*

EXPERIENCE

AI Research Intern Jan 2026 – Jul 2026
CYStar, Department of CSE, IIT Madras

- Developed, trained, and evaluated Machine Learning and Deep Learning models for phishing and brand impersonation detection using Computer Vision and NLP.
- Built end-to-end data pipelines covering ingestion, cleaning, preprocessing, feature extraction, and model evaluation on large-scale real-world image and text data.
- Deployed inference services using FastAPI and Docker, following Git-based version control for reliable, reproducible workflows.
- Collaborated with a research team, benchmarking model performance and communicating findings clearly to technical and non-technical stakeholders.

PROJECTS

UnboxingAI – AI-Powered Learning Companion *React, Express, Gemini, Tavily, SQLite*

- Built a full-stack AI learning platform that ingests live AI news and research, rewriting and adapting each story to the learner's chosen level (Beginner/Intermediate/Expert) using Google Gemini, with Tavily for cross-referencing coverage across outlets.
- Implemented Google OAuth 2.0 (Passport + Express sessions) and a persistent, per-account learner state (saved stories, playlists, streaks, nudge preferences) backed by SQLite.
- Built a React + Vite front-end with a level-adapted story reader, research explorer, and a personal learning dashboard. Live Demo

Purchase Value Prediction – ML Competition *Python, XGBoost, Pandas, Feature Engineering*

- Built a Gradient Boosting (XGBoost) regression model to predict session-level purchase value on a large-scale, 105-column e-commerce dataset.
- Handled missing values, encoded categoricals, and engineered features from session, device, and geographic data to model a highly imbalanced, zero-inflated target.
- Ranked **159th out of 1,800+ participants** (top 9%) with an R^2 of **0.68** on out-of-sample test data.

TDS Virtual TA *Python, FastAPI, RAG, LLMs, FAISS*

- Built a Generative AI assistant using Retrieval-Augmented Generation (RAG): scraped and cleaned Discourse forum data, generated embeddings, and indexed them with FAISS for semantic retrieval.
- Integrated OpenAI's LLM API for response generation, applying prompt engineering to ground answers in retrieved content and reduce hallucinations.
- Deployed scalable REST APIs using FastAPI. Live Demo

Insurance Premium Prediction *Python, Scikit-Learn, Random Forest, XGBoost*

- Built and compared Random Forest and XGBoost regression models to predict insurance premiums, achieving an R^2 of 0.83; deployed via FastAPI with a Streamlit dashboard.

TECHNICAL SKILLS

Programming: Python, SQL, C++, JavaScript
Machine Learning: Scikit-Learn, XGBoost, Random Forest, Regression, Classification, Hyperparameter Tuning, Model Evaluation
Deep Learning: PyTorch, TensorFlow, CNNs, Transfer Learning
Computer Vision: OpenCV, Feature Extraction, Object Detection
NLP & Generative AI: Tokenization, Embeddings, Transformers, LLMs, RAG, Prompt Engineering, Gemini API, Tavily API, OpenAI API
Full-Stack Development: React, Vite, Vue.js, Express, FastAPI, REST APIs, OAuth 2.0
Data Science: Pandas, NumPy, Exploratory Data Analysis, Feature Engineering
Tools & Deployment: Docker, Git/GitHub, Linux, Streamlit
Databases: MySQL, PostgreSQL, SQLite

CERTIFICATIONS

NPTEL Database Management Systems (IIT Madras)
Machine Learning Workshop – IIT Madras
HackerRank SQL (Advanced)